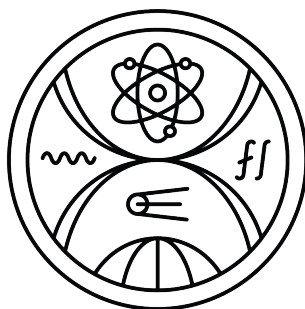


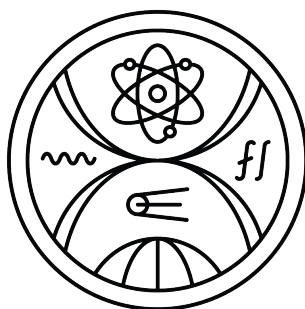
COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



**SIMULATION AND SYNTHETIC DATA
GENERATION OF DEFORMABLE OBJECTS**

Master thesis

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



SIMULATION AND SYNTHETIC DATA GENERATION OF DEFORMABLE OBJECTS

Master thesis

Study program: Applied informatics
Branch of study: Applied informatics
Department: Department of Applied Informatics
Supervisor: doc. RNDr. Martin Madaras, PhD.
Consultant: Mgr. Lukáš Gajdošech, PhD.



Univerzita Komenského v Bratislave
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ZADANIE ZÁVEREČNEJ PRÁCE

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Študijný odbor: informatika
Typ záverečnej práce: diplomová
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Simulation and Synthetic Data Generation of Deformable Objects
Simulácia a generovanie syntetických dát deformovateľných objektov

Anotácia: Tento výskum sa zameriava na generovanie syntetických 3D dát deformovateľných objektov. Na internete je dostupné obrovské množstvo 2D anotovaných datasetov, avšak 3D datasety zamerané na špecifické 3D snímacie zariadenia sú obmedzené. Navyše, v prípade deformovateľných objektov je samotný proces anotácie ťažko vykonateľný na skutočných 3D skenoch a tiež je ťažké ich simulovať a následne renderovať. Príkladom deformovateľných objektov je balenie oblečenia v aplikácii na tzv. bin-picking alebo plastový odpad v aplikácii na triedenie odpadu.

Kľúčové slová: Generovanie syntetických dát, deformovateľné objekty, priehľadné objekty, balenie oblečenia, triedenie odpadu, simulácia, digitálne dvojča

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Dátum zadania: 28.09.2025

Dátum schválenia: 13.10.2025
prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

.....
študent

.....
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THESIS ASSIGNMENT

Name and Surname: Bc. Tomáš Barančok
Study programme: Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: Simulation and Synthetic Data Generation of Deformable Objects

Annotation: This research focuses on generation of synthetic 3D data of deformable objects. An enormous amount of 2D annotated datasets is available over the internet, however 3D datasets focus on a specific 3D capturing hardware are limited. Moreover, in the case of deformable objects, the annotation process itself is hard to perform on real 3D scans and also they are hard to simulate and render. An example of deformable objects is packing of clothing in a bin-picking application or plastic waste in a waste-sorting application.

Keywords: Synthetic data generation, deformable objects, transparent objects, clothing packing, waste sorting, simulation, digital twin

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Guarantor of Study Programme

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Student

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Supervisor

I hereby declare that I have written this thesis by myself, only with help of referenced literature, under the careful supervision of my thesis advisor.

Bratislava, 2026

.....
Bc. Tomáš Barančok

Acknowledgement

Yes... TODO

Abstract

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Keywords: Synthetic data generation, deformable objects, transparent objects, clothing packing, waste sorting, simulation, digital twin

Abstrakt

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Kľúčové slová: Generovanie syntetických dát, deformovateľné objekty, priehľadné objekty, balenie oblečenia, triedenie odpadu, simulácia, digitálne dvojča

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Terminology

Terms

- **one**
One.

Abbreviations

- **ABB** - abbreviation.

Chapter 1

Introduction

This is our work, we do this, we do this because this and these things can be found in our work.

Chapter 2

Overview

2.1 Related Works

These are some works we mention. These include for example [1] and [2]. May include subsections.

2.2 Implementation Problems

Unused engines, ISAAC Sim, ...

Chapter 3

Methodology

3.1 Blender

3.1.1 Rendering Engine

3.1.2 Scripting

3.1.3 Bakes

There will probably be more things, but currently I don't know.

3.2 Physics Simulation

3.3 Realistic metrics

Subject to change.

Chapter 4

Implementation

4.1 Installation and Requirements

4.2 Bottle Parametrization

4.3 Material Simulation and Deformation

Cloth bad, soft-body good, yap yap yap . . .

4.4 Rendering

4.5 Exporting

Chapter 5

Data Generation

5.1 Scene Creation

5.2 Dataset Creation

5.3 Dataset Testing

Chapter 6

Results

Self-explanatory that this will contain the results.

Chapter 7

Conclusion

We conclude our work here.

Bibliography

- [1] Kubirita M. and Gajdošech L. Synthetic dataset rendering with parametric transparent and translucent objects. *Bachelors Thesis*, 2024.
- [2] Paul Schulz, Thorsten Hempel, and Ayoub Al-Hamadi. Automated 3d dataset generation for arbitrary objects. *IEEE Access*, 2025.